knowledge-graph

Knowledge Graph Contract — Extraction, Mentions, Authorship, Pipeline

Feature: `specs/002-knowledge-areas-deep-linking` | **Date**: 2026-09-01

Companion to `http-api-delta.md`, which covers the wire. This document covers **how the graph comes to exist**: minting, promotion, extraction, reconciliation, mention derivation, authorship rules and the repeatable pipeline. Entity shapes live in `../data-model.md` and are not restated. The knowledge-model contract in the workshop's own documentation is authoritative for the ported hierarchy (FR-005); where it and this document describe the same thing, it wins.

1. Minting — normative

- **M1** — every new entity mints through the **existing** passage minter. There is no second minter and no second identifier format.
- **M2** — minting happens **once, at creation**. Re-running any stage over unchanged input mints **zero** identifiers.
- **M3** — an identifier is never derived from the entity's text, title, slug, ordinal, heading, file position or any hash of these.
- **M4** — minting is the **last** branch of every matching algorithm, never the first. If an entity can be matched to an existing one, it is matched; minting is what happens when it demonstrably cannot.
- **M5** — an anchor that is present but unknown to the registry is an **error**. It is never a silent mint and never an adoption. This mirrors the passage registry's existing hard edge, and it exists because the alternative — adopt on mismatch — turns a corrupted reference into a plausible one.

Gate G-KG-13 — run every stage twice over unchanged input and assert zero mints, a byte-identical taxonomy file, and no source file modified. **Paired mutation**: make minting unconditional whenever no anchor was read in the current run; the gate must go red on all three assertions at once.

1.1 `screen_text` — a fifth kind, not a fifth registry

- **M6** — on-screen text recognised from a chapter recording enters the corpus as a **passage** of kind `screen_text` (FR-060, data-model §2.9). It mints through the **same** minter, resolves through the **same** four-outcome resolver, and inherits the **same** append-only redaction log. There is no second registry, no second mint path and no second identifier format.
- **M7** — its ordering key is the **visibility onset**, exactly as a transcript segment's is its start time, so spoken and on-screen passages of one chapter order on **one** axis.
- **M8** — it declares `producer = ocr`, **distinct from `asr`**, and that marking is enforced in **both** directions: only a `screen_text` may claim it, and a `screen_text` may claim nothing else. This is **not** the library's `provenance` field, which records whether the text is still the machine's or has been human-corrected — a corrected OCR passage is `human_corrected` **and** `ocr`, and collapsing the two axes would make that state unrepresentable.
- **M9** — it carries the engine's own **confidence** and the **interval bound** (FR-061). The bound is never omitted and never zero: an interval read off a sampling grid is not exact, so a zero bound is not a tight measurement but a measurement never taken, presented as the tightest possible one.
- **M10** — an **empty recognition is not a passage**. It is a sample that recognised nothing and is reported as such, so evidence counts are never inflated by rows that evidence nothing.

Gate G-OCR-2 — assert a `screen_text` mints through the existing minter into the existing registry, resolves to all **four** outcomes, orders on its visibility onset, carries producer, confidence and interval bound through the existing wire translation with no per-kind branch, and is suppressed by the existing redaction log alongside a spoken passage. **Paired mutation**: mint an OCR passage through a second identifier format, or strip any one of the invariants above; the gate must go red, and it must go red **naming the invariant that broke** — a refusal from an unrelated rule is a catch the gate did not make.

`screen_text` is deliberately NOT advertised as a retrievable kind. Nothing produces one yet (002 T126 is unbuilt and blocked). `diagram` is already an advertised-but-unretrievable kind in this deployment, and adding a second instance of that defect to buy a tidier-looking kind list is refused.

2. Promotion, extraction, reconciliation — the order is normative

Order per research D-KG-3. Running these out of order produces a defensible-looking result that has quietly inverted which source of truth holds the identifiers.

2.1 Promote (first)

- **P1** — each area that exists today as prose becomes a first-class entity with a minted identifier.
- **P2** — each promoted area is evidenced against the corpus. **An area that acquires zero evidencing mentions fails promotion loudly** and is not published. It is not published unevidenced and it is not silently dropped; both are reported.
- **P3** — the prose document becomes the area's materials, and its existing section headings become lesson sections. It is not rewritten as a side effect of promotion.
- **P4** — promotion records, per area, that its origin is `promoted`. This is not decoration: it tells a later reader which areas embody human judgement about what the workshop teaches and which a machine proposed.

2.2 Extract (second)

- **E1** — extraction reads the **whole corpus** and proposes areas and terms. It never reads the reference module (FR-004a).
- **E2** — every proposal carries its evidence and its significance measure **with the inputs to that measure**, so a reviewer can disagree with a specific term and see why it scored as it did.
- **E3** — raw frequency alone is **not** a significance measure (FR-013). A measure that ranks a spoken corpus by frequency returns function words.
- **E4** — a proposal with a low-confidence score is marked `uncertain` and is never presented as confident (FR-011).

- **E5** — **no area count is targeted.** The corpus evidences as many areas as it evidences. Producing areas to match the reference's 34 would mean inventing coverage the recording does not contain, which is the failure this whole contract is arranged against. Research **U3** records that the actual number is unknown until the first run, and **no downstream artifact may assume one**.

2.3 Reconcile (third, and visibly)

Every extracted proposal falls into exactly one of three branches, and the third is the one that matters:

- **R1 — attaches**: the proposal is within a promoted area's scope. It becomes evidence on that area. No new area, no new identifier.
- **R2 — adds**: the proposal is outside every promoted area's scope. A new area is minted with origin `extracted`.
- **R3 — contradicts**: the proposal overlaps a promoted area but disagrees about its scope or boundary. It is **reported for a decision**. It is **neither merged nor discarded**, and the pipeline does not proceed past it silently.

Why R3 is a hard stop rather than a heuristic: merging on overlap silently redraws a boundary a human drew. Discarding on overlap silently ignores evidence. Either is a decision about what the workshop teaches, made by a similarity score, invisibly. Reporting costs a review; the alternatives cost the taxonomy's credibility.

Gate G-KG-14 — seed a contradiction, run reconciliation, and assert it appears in the report and that neither a merge nor a discard occurred. **Paired mutation**: make R3 fall through to R1 above a similarity threshold; the gate must go red.

2.4 Account for everything

- **A1** — every passage is either attached to at least one area or **explicitly classified as unattached with a stated reason** (FR-010).
- **A2** — attached plus classified equals the registry count. Exactly. A shortfall is a failure, not a rounding.
- **A3** — the **proportion** attached is published as a measured figure. **No threshold is set** (research U2): the density of teachable substance in a spoken working session has not been measured, and a guessed target gets optimised toward.

3. Mention derivation — the measured join

The algorithm is normative because its two branches produce genuinely different guarantees, and collapsing them is invisible.

Given an occurrence of a term or area in a passage:

1. Record the **text span** — character offsets within the passage's text. Always.
2. If the passage is **not** a transcript segment, stop. No time span, no precision.
3. If it **is**, attempt the word join: locate the occurrence's words in the chapter's word sidecar and take their start and end times.
    - **Joined** → time span from the word times; **precision** = **`word`**; carry the word's timing confidence.
    - **Not joined** → time span from the enclosing segment; **precision** = **`segment`**.
4. If the occurrence crosses a segment boundary, emit **one mention per segment it touches** (FR-022).

Measured, and both branches are real: 15,610 words, monotonic, spanning exactly the same range as the segment set. **15,535 (99.52%) join to a segment by time; 75 (0.48%) do not**, falling in the gaps between segments. The second branch is not a theoretical fallback.

- **N1** — `precision` is **required** whenever a time span exists. There is no default. A default would make every unjoined mention claim word accuracy.
- **N2** — the two precisions are **never** conflated in storage, on the wire, or in the interface. Measured segment durations are median 6.74 s, p95 10.78 s and maximum 20.22 s, so the difference between the branches is up to twenty seconds of recording — long enough to be a different topic.
- **N3** — timing confidence is carried, not discarded. 17.4% of words fall below the recorded flag threshold and the 5th percentile is 0.559; "usually confident" is the reasoning that ships the wrong link.
- **N4** — the join is by **time into segment**, and the segment yields the **passage identifier**. The identifier is what is stored. The time is a position (FR-033b).

Gate G-KG-4 (shared with the wire contract) — assert every time-carrying mention declares its precision. **Paired mutation**: omit precision and default it to `word`; the gate must go red.

Gate G-KG-15 — assert the unjoined fraction is measured and reported rather than assumed zero. **Paired mutation**: silently treat an unjoined occurrence as `word` precision at the segment start; the gate must go red. Research **U5** asks whether the 75 unjoined words cluster or scatter; the answer changes nothing about this contract but changes whether there is a segmentation defect worth fixing.

4. Authorship — materials and questions

4.1 Provenance of prose

- **W1** — every lesson section records `authored` or `assembled` (FR-017). Under a purely extractive authorship decision everything is `assembled`; under an agent-authored one everything is `authored`; under a generated one both appear. The field is built now so all three remain shippable without a schema change.
- **W2** — every substantive claim carries at least one resolving citation, or is **visibly marked as editorial framing that is not workshop content** (FR-015). There is no unmarked, uncited claim.
- **W3** — an area is not published until its publication review is recorded, and a review older than the materials it reviews is **stale** and fails (FR-016). Recording *"nothing to change"* is a valid review; skipping it is not.

4.2 Questions

- **Q1** — every question carries at least one citation and **every citation must resolve**, or the question is **withheld** (FR-035, FR-036). Withholding is the default path, not an error path.
- **Q2** — a `long`-set question cites **more than one distinct** passage; a `short`-set question may cite one (D-KG-7). This is the structural distinction, chosen because perceived difficulty is not reproducible and citation breadth is.
- **Q3** — every question names the lesson sections it assesses. This is what makes per-section coverage measurable rather than estimated.
- **Q4** — an assembled model answer is marked assembled wherever it appears (FR-038).
- **Q5** — **no question, prompt, choice, answer or explanation may originate from the reference module** (FR-004, FR-004a). It holds 785 finished questions about a different subject, in a private

repository. Reusing them would be faster, would survive a superficial review, and would be worthless.

Gate G-KG-16 — plant one reference-module question in a workshop bank and assert the content boundary check fails. **Paired mutation**: scope the boundary check to the outbound direction only; the gate must go red. The check runs **both** ways deliberately: outbound keeps a private recording out of public repositories, inbound keeps the workshop from shipping someone else's curriculum under its own name.

4.3 Coverage measurement

- **C1** — every lesson section of every published area is assessed by at least one question (SC-017b). A section with zero is listed explicitly, never omitted from the enumeration.
- **C2** — the proportion of an area's evidencing passages cited by at least one question is computed **per area** and published as a **distribution**, never as a mean (SC-017c). A mean is how a well-covered area conceals an empty one.
- **C3** — **no threshold** is expressed and no field says "pass" (research U2).

5. The repeatable pipeline

The feature is complete when the **next** chapter runs through unchanged, not when this chapter's materials look finished.

- **S1** — the knowledge layer is a set of stages **added to the platform's existing chapter-addition path** — its documented procedure and its prompt — not a second way to add a chapter (FR-033e). Two procedures for one act guarantee one of them rots.
- **S2** — processing a chapter yields, without hand-authoring of structure: transcript, passages with minted identifiers, extracted areas and themes, authored materials, short and long question sets, cross-references, index entries, deep links.
- **S3** — every stage is **three-valued**: determined-good, a real problem found, or could not determine. A missing dependency, an unreachable service or a crashed helper is always the third.
- **S4** — every stage is **idempotent** (M2) and **resumable**, and reports progress from measured rate rather than from an estimate.

- **S5** — a chapter that cannot be fully processed reports **precisely what is missing** and publishes nothing (FR-033g). A partially processed chapter is never presented as complete.
- **S6** — adding a chapter **updates** the taxonomy: established areas keep their identifiers and gain evidence, newly evidenced areas are added, contradictions are reported (FR-033f, §2.3 R3).
- **S7** — no stage writes to a source. Sources are opened read-only; derived artifacts land only in the derived locations.
- **S8** — every stage writes its evidence to the versioned evidence location, including — and especially — on a could-not-determine outcome, which is the run a reader most needs the record of.

Gate G-KG-17 — run the whole pipeline against a small synthetic chapter and assert every output in S2 exists and that the diff contains **no hand-created structural file**. **Paired mutation**: remove one stage from the procedure; the gate must go red naming the missing output. This is the gate that proves the pipeline is repeatable rather than that one chapter happens to be finished.

Gate G-KG-18 — withhold one required input and assert the run names exactly what is missing and publishes nothing. **Paired mutation**: downgrade the missing input to a warning; the gate must go red.

6. Resolution and redaction

Both are inherited from the platform's existing contracts and are **not** reimplemented here. Two consequences bind this feature specifically:

- **X1** — every link, citation and mention resolves through the **one** existing resolution function, with its four outcomes. There is no second path and no fallback of any kind — no fuzzy text match, no nearest neighbour, no prefix match, no same-hash lookup (FR-023, FR-024).
- **X2** — redaction propagates to all **eight** targets enumerated in the data model: mentions, terms, areas, questions, lesson sections, index entries, exports and stored answers. **Gate G-KG-7** skips exactly one and must go red.

7. Environment adaptability

Requirements that follow from measurement on this host, not from principle:

- **V1** — every export and diagram tool is **detected and capability-probed**, never assumed from a name on `PATH`. Measured: all of them resolve into **user-local** directories rather than system ones. This repository has already shipped a defect of exactly this shape — a media tool that answered a version query and rejected the flag that mattered, because the name resolved into an unrelated cache.
- **V2** — word-timing availability is detected **per chapter**, not assumed corpus-wide. One chapter has a word sidecar; nothing guarantees the next one will.
- **V3** — the significance measure prints the host facts and the arithmetic it used, so the score can be audited rather than trusted.
- **V4** — no stage assumes a generative model. There is none on this host, and a stage that requires one reports could-not-determine rather than failing as though the content were at fault.

8. Gate inventory

Gate	Subject	Owned by
G-KG-1	route manifest completeness	wire contract §2
G-KG-2	unprovenanced question withheld	wire §3.5 / here §4.2 Q1
G-KG-3	long-set citation breadth	here §4.2 Q2
G-KG-4	precision declared on every time-carrying entry	here §3
G-KG-5	offsets populated or field absent	wire §4.1
G-KG-6	no time-keyed relationship	here §3 N4
G-KG-7	redaction reaches all eight targets	here §6 X2
G-KG-8	unresolvable link fails loudly	here §6 X1

Gate	Subject	Owned by
G-KG-9	export toolchain unavailable ⇒ could-not-determine	here §7 V1
G-KG-10	unevidenced area or term never served	here §2.1 P2
G-KG-11	advertised kind is actually retrievable	wire §5
G-KG-12	coverage per area, never a mean	here §4.3 C2
G-KG-13	idempotency: zero mints on re-run	here §1
G-KG-14	contradiction reported, never merged or discarded	here §2.3
G-KG-15	unjoined mention fraction measured, not assumed zero	here §3
G-KG-16	content boundary enforced **in both directions**	here §4.2 Q5
G-KG-17	synthetic chapter produces every output, no hand-assembly	here §5
G-KG-18	incomplete chapter names what is missing, publishes nothing	here §5
G-OCR-2	`screen_text` is a fifth kind, not a fifth registry	here §1.1

Nineteen gates, nineteen paired mutations. A gate that has never been observed failing is not known to work, and this repository has twice shipped a proof that could not fail — one whose control failed so zero mutations ever ran, one that exercised only sandboxed copies while the real entry point could not start. **Every proof here must include a case that runs the real entry point against the real tree.**

G-OCR-2 is the only Phase-11 gate in this table, and the omission of the rest is deliberate. Phase 11 names eleven `G-OCR-*` identifiers in its task list. Ten of them guard code nobody has written, and publishing an identifier here before its gate exists inverts the order this contract depends on: the closure check requires every gate id in `contracts/` to be carried by a task block, and the registry requires every registered check to have a paired demonstration. An id published ahead of its gate satisfies both bookkeeping rules while asserting a demonstration that does not exist — which is the inoperative-proof defect wearing a contract's clothes. **Build the gate, then add the row.**